ASSIGNMENT - 9

① The kinetic energy of a particle :

$$T = (\gamma - 1) mc^2 \quad \text{—— ①}$$

The work done on a particle due to a constant force :

$$W = Fx \quad \text{—— ⑪}$$

Work done on a particle = Change in kinetic energy

$$\therefore ① = ⑪ \rightarrow \quad Fx = (\gamma - 1) mc^2$$

$$\Rightarrow \quad \gamma - 1 = \frac{Fx}{mc^2} = \frac{x}{\varepsilon}$$

$$\Rightarrow \quad \gamma = \frac{\varepsilon + x}{\varepsilon}$$

$$\Rightarrow \quad \left(1 - \frac{u^2}{c^2}\right)^{1/2} = \frac{\varepsilon}{\varepsilon + x}$$

$$\Rightarrow \quad 1 - \frac{u^2}{c^2} = \frac{\varepsilon^2}{(x+\varepsilon)^2}$$

$$\Rightarrow \quad u^2 = \left[\frac{(x+\varepsilon)^2 - \varepsilon^2}{(x+\varepsilon)^2}\right] c^2$$

$$\Rightarrow \quad \boxed{u = \sqrt{\frac{x(x+2\varepsilon)}{(x+\varepsilon)}}\, c}$$

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(2) $p_\gamma = p'_\gamma \cos\theta + p_e' \cos\phi$ — (I)

$p_\gamma' \sin\theta = p_e' \sin\phi$ — (II)

$\Rightarrow \sin\phi = \dfrac{p'_\gamma \sin\theta}{p_e'}$ — (III)

$\Rightarrow \sqrt{1 - \cos^2\phi} =$

$\left(\dfrac{p_\gamma - p_\gamma' \cos\theta}{p_e'}\right) = \cos\phi$ — (IV)

$(III)^2 + (IV)^2 \Rightarrow 1 = \left(\dfrac{p_\gamma' \sin\theta}{p_e'}\right)^2 + \left(\dfrac{p_\gamma - p_\gamma' \cos\theta}{p_e'}\right)^2$

$= \dfrac{1}{p_e'^2}\left[p_\gamma'^2 \sin^2\theta + p_\gamma^2 + p_\gamma'^2 \cos^2\theta - 2p_\gamma p_\gamma' \cos\theta\right]$

$\cancel{= \dfrac{1}{p_e'^2}\left[p\ 2p_\gamma^2 - 2p_\gamma p_\gamma' \cos\theta\right]}$

$\cancel{= \dfrac{2p_\gamma}{p_e'^2}\left[p_\gamma - p_\gamma' \cos\theta\right]}$

$\Rightarrow p_e'^2 = p_\gamma'^2 \cancel{\sin^2\theta} + p_\gamma^2 + \cancel{p_\gamma'^2 \cos^2\theta} - 2p_\gamma p_\gamma' \cos\theta$ — (V)

Relativistic relatn b/w energy & momentum:

$$E_e' = \left(p_e'^2 c^2 + m_e^2 c^4\right)^{1/2}$$

$\Rightarrow m_e c^2 + p_\gamma c - p_\gamma' c = \left(p_e'^2 c^2 + m_e^2 c^4\right)^{1/2}$ (from (I))

$\Rightarrow \cancel{m_e^2 c^4} + c^2 (p_\gamma - p_\gamma')^2 + 2m_e c^3 (p_\gamma - p_\gamma') = \cancel{m_e^2 c^4} + p_e'^2 c^2$

$\Rightarrow p_e'^2 = (p_\gamma - p_\gamma')^2 + 2m_e c (p_\gamma - p_\gamma')$ — (VI)

Using (vi) in (v):

$$(p_\gamma - p_\gamma')^2 + 2m_e c(p_\gamma - p_\gamma') = p_\gamma'^2 \cancel{\sin^2\theta} + p_\gamma^2 + \cancel{p_\gamma^2 \cos^2\theta} - 2p_\gamma p_\gamma' \cos\theta$$

$$\Rightarrow \cancel{p_\gamma^2} + \cancel{p_\gamma'^2} - 2p_\gamma p_\gamma' + 2m_e c(p_\gamma - p_\gamma') = \cancel{p_\gamma'^2 \sin^2\theta} + \cancel{p_\gamma^2} + \cancel{p_\gamma^2 \cos^2\theta} - 2p_\gamma p_\gamma' \cos\theta$$

$$\cancel{\Rightarrow p_\gamma'^2 \cos^2\theta - 2p_\gamma p_\gamma' + 2m_e c(p_\gamma - p_\gamma') = p_\gamma^2 \cos^2\theta - 2p_\gamma p_\gamma' \cos\theta}$$

$$\cancel{\Rightarrow \cos^2\theta (p_\gamma'^2 - p_\gamma^2) = 2p_\gamma p_\gamma' (1 - \cos\theta) - 2m_e c(p_\gamma - p_\gamma')}$$

$$\Rightarrow \cancel{2}p_\gamma p_\gamma' (1 - \cos\theta) = \cancel{2}m_e c(p_\gamma - p_\gamma')$$

$$\Rightarrow \frac{h}{\lambda} \cdot \frac{\cancel{h}}{\lambda'} (1 - \cos\theta) = m_e c \left(\frac{\cancel{h}}{\lambda} - \frac{\cancel{h}}{\lambda'}\right)$$

$$\Rightarrow h \frac{(1 - \cos\theta)}{\cancel{\lambda\lambda'}} = m_e c \left(\frac{\lambda' - \lambda}{\cancel{\lambda\lambda'}}\right)$$

$$\Rightarrow \boxed{\lambda - \lambda' = \frac{h}{m_e c}(1 - \cos\theta)}$$

(3) Initial state:

Photon 4-momentum:

$$P_\gamma = (|p|, p) \quad \text{— (1)}$$

where $|p|$ is the energy of the photon.
($\because$ a photon is massless, energy $|p| = c|p|$ is natural, where $c = 1$)

Electron 4-momentum (at rest initially):

Orion

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$$p_e = (m, 0) \quad \text{—} \quad (II)$$

where m is the rest mass of the e^-.

Final state:

After the photon is absorbed, the e^- gains all the energy and momentum of the photon:

$$p_e' = p_\gamma + p_e = \left(\underbrace{|p| + m, p}_{\sqrt{(|p| + m)^2 - |p|^2}}\right) \quad \text{—} \quad (III)$$

For energy-momentum conservation, the final state of the e^- must satisfy:

$$p_e'^2 = p_e' \cdot p_e' = m^2 \quad \text{—} \quad (IV)$$

where $p_e'^2$ is the square of the 4 momentum of the e^- and it must equal m^2, the invariant mass of the e^-

from eq^n (III) & (IV):

$$(|p| + m)^2 - |p|^2 = m^2$$

$$\Rightarrow \cancel{|p|^2} + \cancel{m^2} - 2m|p| - \cancel{|p|^2} = \cancel{m^2}$$

$$\Rightarrow 2m|p| = 0$$

$\because m \neq 0$; $|p| = 0$.

This means the photon has zero energy, which is unphysical. Therefore the process cannot conserve energy & momentum simultaneously.

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(4) The isotope Francium-220 (^{220}Fr) decays via 2 processes

(i) α-decay (main mode)

$$^{220}Fr \longrightarrow {}^{216}At + \alpha$$

The decay releases an energy of approximately 6-8 MeV carried predominantly by the α particle

(ii) β-decay (rare mode):

$$^{220}Fr \longrightarrow {}^{220}Ra + \beta^-$$

The maximum energy of the β particle emitted is approximately 1.21 MeV

For the α-decay, the α particle (4 amu) gets the majority of the decay energy due to its much smaller m compared to the recoiling nucleus (216 amu). Using momentum conservation:

$$K_\alpha = 6.8 \times \frac{216}{220} \text{ MeV} = 6.67 \text{ MeV}$$

For the β decay, the energy released (1.21 MeV) is shared between the β particle & the recoiling nucleus, so the β particle's maximum K.E. is 1.21 MeV.

(5) 4-momentum of a particle:

$$\mathbf{p} = \left(\frac{E}{c}, \vec{p}\right)$$

where E is the total energy and $\vec{p}$ is the 3-momentum

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For the 1st particle in Alice's Frame:

$$P_1 = (\gamma_1 m_1 c, \gamma_1 m_1 \vec{u_1}) \; ; \; \gamma_1 = \frac{1}{\sqrt{1-u_1^2/c^2}}$$

For the 2nd particle (and Bob, who moves with the same velocity $\vec{u_2}$ as the 2nd particle relative to Alice):

$$P_2 = (\gamma_2 m_2 c, \gamma_2 m_2 \vec{u_2}) \; ; \; \gamma_2 = \frac{1}{\sqrt{1-u_2^2/c^2}}$$

The total energy of the 1st particle as observed by Bob is:

$$E_1' = \gamma_{12} m_1 c^2 \; ;$$

γ_{12} – Lorentz factor for the relative velocity $\vec{u_{12}}$ of the 1st particle w.r.t. Bob.

$$P_1 \cdot P_2 = P_1' \cdot P_2' \quad \text{(invariant)}$$

Alice's Frame: $P_1 \cdot P_2 = \frac{E_1 E_2}{c^2} - \vec{p_1} \cdot \vec{p_2}$ $\left[E_1 = \gamma_1 m_1 c^2 ; \; E_2 = \gamma_2 m_2 c^2 \right]$

Bob's Frame: $P_1' \cdot P_2' = \frac{E_1' m_2 c^2}{c^2}$

Equating the two: $\frac{E_1' m_2 c^2}{c^2} = \frac{E_1 E_2}{c^2} - \vec{p_1} \cdot \vec{p_2}$

$$= \gamma_1 \gamma_2 m_1 m_2 (c^2 - \vec{u_1} \cdot \vec{u_2})$$

$$\Rightarrow \boxed{E_1' = \gamma_1 \gamma_2 m_1 m_2 (c^2 - \vec{u_1} \cdot \vec{u_2})}$$